To understand the next problem we now define, **The Rank of a word in the dictionary.** It is the place at which the given word comes when writing all the strings formed by the letters of the given word in the dictionary order or lexicographic order.

Example 4.35 If the letters of the word TABLE are permuted in all possible ways and the words thus formed are arranged in the dictionary order (alphabetical order), find the ranks of the words (i) TABLE, (ii) BLEAT

Solution:
The dictionary order of the letters of given word is A, B, E, L, T. In the dictionary order of the words which begin with A come first. If we fill the first place with A, remaining 4 letters (B, E, L, T) can be arranged in $4!$ ways. On proceeding like this we get

(i) The rank of the word TABLE

$A - - - - = 4! = 24$ ways
$B - - - - = 4! = 24$ ways
$E - - - - = 4! = 24$ ways
$L - - - - = 4! = 24$ ways
$TABEL = 1$ way
$TABLE = 1$ way

The rank of the word TABLE is $4 \times 4! + 1 + 1 = 98$.

(ii) The rank of the word BLEAT

$A - - - - = 4! = 24$ ways
$BA - - - = 3! = 6$ ways
$BE - - - = 3! = 6$ ways
$BLA - - = 2! = 2$ ways
$BLEAT = 1$ way

The rank of the word BLEAT is $24 + 6 + 6 + 2 + 1 = 39$.

4.4.5 Permutations of not all distinct objects

Consider permuting the letters of the word JEE. In this case the letters of the word are not different. There are 2 E's, which are of same kind. Let us treat, temporarily, the 2 E's as different, say E_1 and E_2. The number of permutations of 3 different letters taken all at a time is $3!$.

Permutations when E_1, E_2 are different	Permutations when E_1, E_2 are the same.
JE_1E_2, JE_2E_1	JEE
E_1JE_2, E_2JE_1	EJE
E_1E_2J, E_2E_1J	EEJ